

① SOLVE THE FOLLOWING SYSTEMS OF EQUATIONS GRAPHICALLY,

$$\begin{cases} \frac{x^2}{4} + y^2 = 1 \\ x^2 - y^2 = 1 \end{cases}$$

② SOLVE THE FOLLOWING SYSTEMS OF EQUATIONS ALGEBRAICALLY.

$$\begin{cases} S = t + 6 \\ \sqrt{S} - \sqrt{t} = 1 \end{cases}$$

③ SOLVE FOR x & y ALGEBRAICALLY

$$\begin{cases} \frac{6}{x} + \frac{3}{y} = 4 \\ \frac{36}{x^2} + \frac{36}{y^2} = 13 \end{cases}$$

④ SOLVE THE EQUATION $e^x + e^{-x} = 2$

⑤ SOLVE FOR x , $\left(\frac{x}{x+1}\right)^2 + \frac{2x}{x+1} = 8$

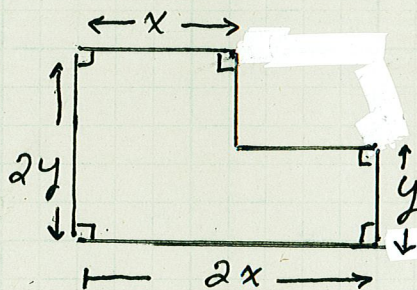
⑥ GIVEN $\sqrt{10 + 3\sqrt{x}} = \sqrt{x}$, SOLVE FOR x

⑦ SOLVE FOR x , if $\log(\sqrt{x} + 38) - \log x = 1$

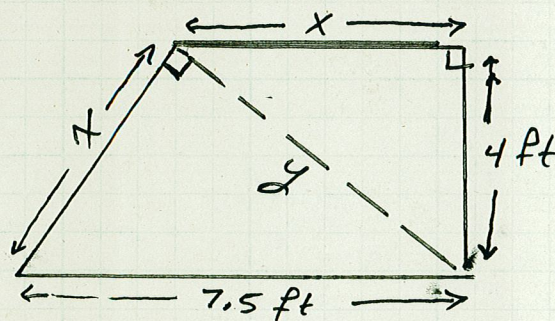
⑧ if $f(x) = \sqrt{8-2x}$ AND $f(x+1) = 2$, FIND x

- (2)
- ⑨ A WRENCH IS DROPPED BY A WORKER AT A CONSTRUCTION SITE, FOUR SECONDS LATER THE WORKER HEARS IT HIT THE GROUND. How HIGH IS THE WORKER ABOVE THE GROUND? (THE VELOCITY OF SOUND IS 1100 ft/s AND THE DISTANCE THE WRENCH FALLS AS A FUNCTION OF TIME IS $S = 16t^2$),

- ⑩ A CIRCUIT ON A COMPUTER CHIP IS DESIGNED TO BE WITHIN THE AREA SHOWN BELOW. IF THIS PART OF THE CHIP HAS AN AREA OF 9.0 mm^2 AND PERIMETER OF 16 mm , FIND x AND y .



- ⑪ FOR THE PLYWOOD PIECE SHOWN, FIND x & y .



- ⑫ A PLASTIC BAND 19.0 cm LONG IS BENT INTO THE SHAPE OF A TRIANGLE WITH SIDE $\sqrt{x-1}$, $\sqrt{5x-1}$ AND 9 . FIND x .